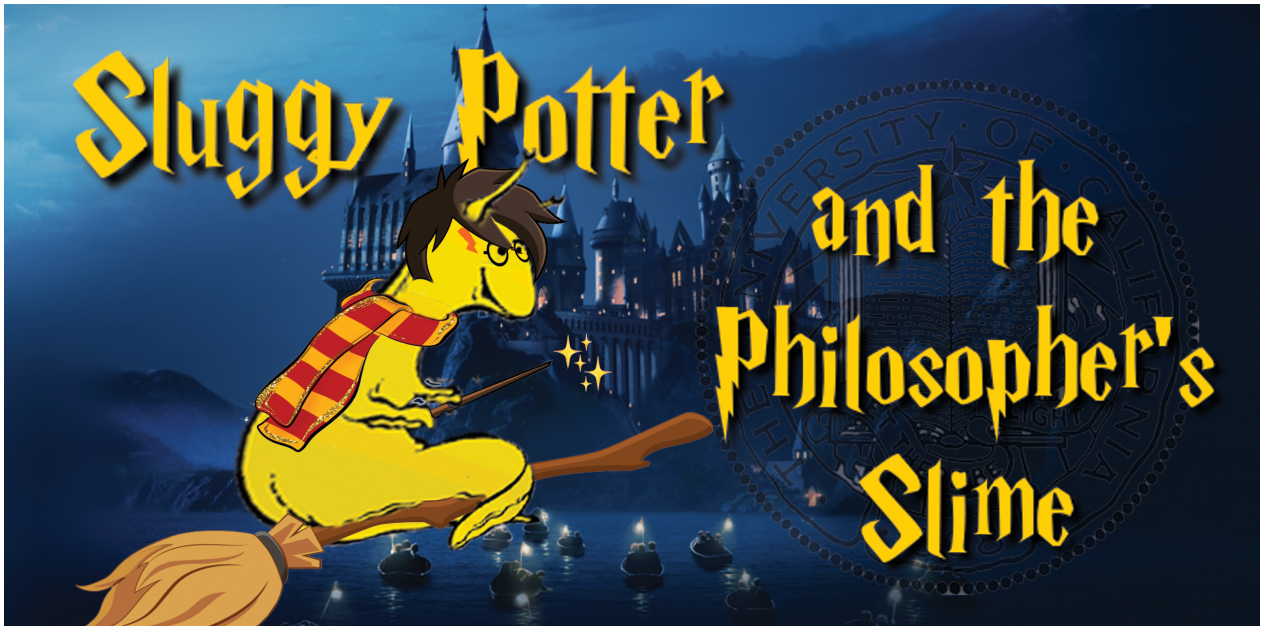
ECE-118/218 Introduction to Mechatronics

Sluggy Potter
and the Philosopher's Slime

University of California Santa Cruz — Fall 2026



Background Briefing

Your magical school for witchcraft and wizardry is hosting this year's *Sluggy Potter and the Philosopher's Slime* competition. Your team has survived the preliminary trials (the labs), and now faces the final challenge: bringing together everything you have learned to build a robotic champion (a fully autonomous robot) worthy of the Triwizard Cup.

Though Muggles have rarely been invited to compete, the ECE-118 class has been granted special permission to enter using the magic of mechatronics. Your spells will be written in C, your charms will take the form of sensors and actuators, and your champion will have to fend for itself. Once the round begins, no helpful incantations from the sidelines will save it.

Your robotic champion will begin somewhere in the start zone. It must find its bearings, locate the opposing robot, and strike it twice with magic orbs (ping-pong balls). But dueling alone is not enough: your champion must also put at least one orb through the designated hole in at least two slime tower targets. Track wire identifies the target, and a vertical banner marks the correct hole. Finding the tower is only half the spell; your robot must recognize the marked opening and aim accordingly.

To complete the challenge, your champion must accomplish both tasks: two hits on the opponent and at least one ball through the designated hole in a minimum of two slime targets. In the tome below, you will find the specifications for your robotic champion and the rules of the field. Gather your teammates, consult your blueprints, and prepare to turn a collection of decidedly non-magical parts into a champion. The Philosopher's Slime awaits.

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Purpose

The purpose of this project is to provide an opportunity to apply all that you have learned in ECE-118/218 to solve an open-ended problem. Your task is to build an autonomous robotic champion that starts near the center of the field in a random orientation, locates the opponent robot, and hits its body twice with ping-pong balls. Your robot must also locate the slime tower target designated by track wire, identify the correct hole by its banner, and put at least one ping-pong ball through that hole on at least two of the three targets. Both tasks are required to complete the challenge. In the public tournament, winners will be determined by points.

Project Requirements

- Team and robot meet three Design Reviews:
  1. Preliminary Design Review,
  2. Mid-Project Review, and
  3. Final (min-spec) Check-Off.
- Team maintains an active lab notebook or website detailing progress and designs.
- Each and every week, the team satisfies check-offs and meets with their tutor.
- All loaned parts are returned to TAs after the tournament.
- Lab is cleaned up before the end of finals week.
- Final Report is due electronically on Thursday, 10-Dec-2026.
- Participation in the Public Tournament on Friday, 04-Dec-2026 (0% of your grade, 100% fun).
- If your robot can demonstrate robust Final Check-Off one week before the final deadline, The Prof. will personally buy your team beer. The beer check-off has rarely been awarded.

Project Overview

Your task is to build a small autonomous robotic champion that can find its bearings, navigate the field, locate the opponent, and aim and launch ping-pong balls accurately. Completing the challenge requires two hits to the opponent robot body and at least one ball through the designated hole in the slime tower target (you must hit at least two of the three slime tower targets). You will be doing this in teams of three, over the next five weeks, during which time you will design, implement, test, and iterate until you can reliably complete both tasks. There will be practice fields (quarter fields) in the labs, and lots of help and guidance available to you. Don't panic. Yet.

The field of play is a large white 8 ft. x 8 ft. surface with a 2 in. black tape boundary, as shown in Fig. 1. Your robot starts somewhere near the center of the field, in a random orientation. It must determine how to reach suitable shooting positions for both the opponent and the slime tower targets.

The opponent robot is identified by its 2 KHz IR beacon. Both your robot and the opponent carry a beacon that remains illuminated throughout the round. Each robot must provide a mounting bracket and a standard power connector from the power distribution board for its beacon assembly. Beacon mounting details will be provided with the project materials.

The slime tower target is designated by track wire operating at the standard 24–26 KHz, and will likely be radiating IR in the 1.5 and/or 2.5KHz range. A vertical banner hanging below the hole identifies the correct hole in that target.* Your robot must use these distinct cues to locate the target and select the correct opening;

*We haven't yet decided if the hanging banner is black, white, or reflective, but you will be able to tell its presence with your standard sensors.

a ball through an unmarked hole does not satisfy the tower task. Target dimensions, hole geometry, track wire routing, and placement details will be provided in the project field drawings.

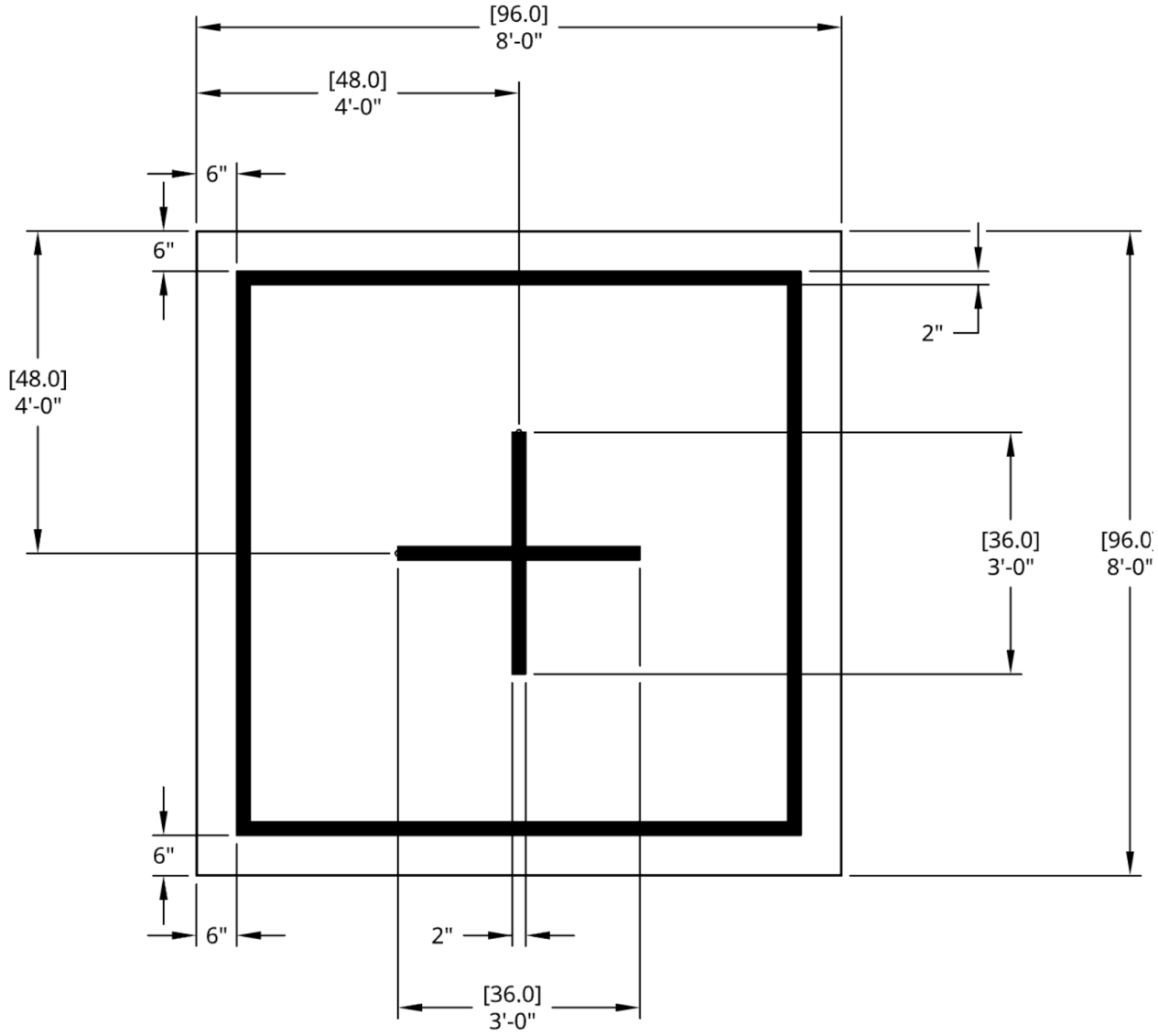


Figure 1: Field of play for Sluggy Potter and the Philosopher's Slime. The field is 8 ft. x 8 ft., with 2 in. wide black tape boundaries and interior alignment marks. Robots start near the center in a random orientation. Opponent and slime tower target placements will be specified separately.

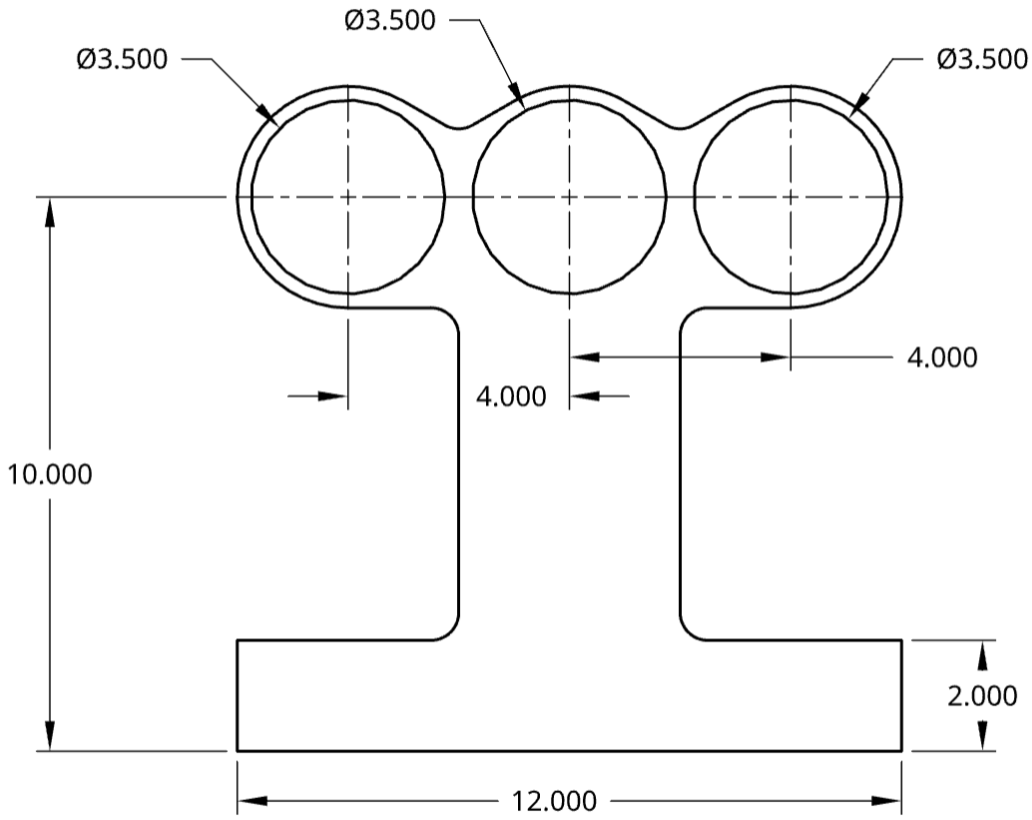


Figure 2: Slime tower target geometry (dimensions in inches). Track wire is located 2" above the ground and runs horizontally, and a vertical banner marks the correct hole. At least one ping-pong ball must pass through that hole. Note that the slime tower will likely radiate IR at 1.5 and/or 2.5KHz.

Each robot must start the match within an 11 in. x 11 in. x 11 in. cube volume. Parts may move after the round begins, but the robot must remain intact throughout the match. Jamming your opponent in any way is disallowed. Robot sizing will be checked with the *Cube of Compliance*.[†]

Your robot must remain within the field, with at least half of the robot inside the black tape boundary at all times. If more than half of the robot is past the boundary, it will be disqualified.[‡] Your robot must detect collisions and break contact within 5 seconds or be disqualified. It must be able to maneuver around any immovable object blocking its path.

While every attempt has been made to finalize the project specifications and rules, understand that this is a work in progress. As the project evolves, we will make minor tweaks to the specs as we discover flaws that we have not anticipated. These will be announced in class and posted on Ed Discussions. They are not meant to destroy your winning design, but only to make things work smoothly. Your understanding is appreciated.

[†]We have (and will) require you to modify your robot because it does not fit inside the dreaded Cube of Compliance. Remember to take into account things that stick off your robot (e.g.: wires).

[‡]This is for your own protection: the field is high off the ground, and several robots have tried driving off the table before.

Minimum Specification Checkoff

In order to pass this class, your robot must demonstrate that it can complete both tasks.[§] Teams are free to embellish, go beyond, and otherwise have fun. However, we suggest you aim for "min spec" first, and then go back and go nuts.[¶]

Your robot begins the round somewhere near the center of the playing field, in a random orientation. The opponent robot and slime tower targets are placed in valid locations specified by the checkoff field configuration. Your robot must autonomously find its bearings and navigate to suitable shooting positions.

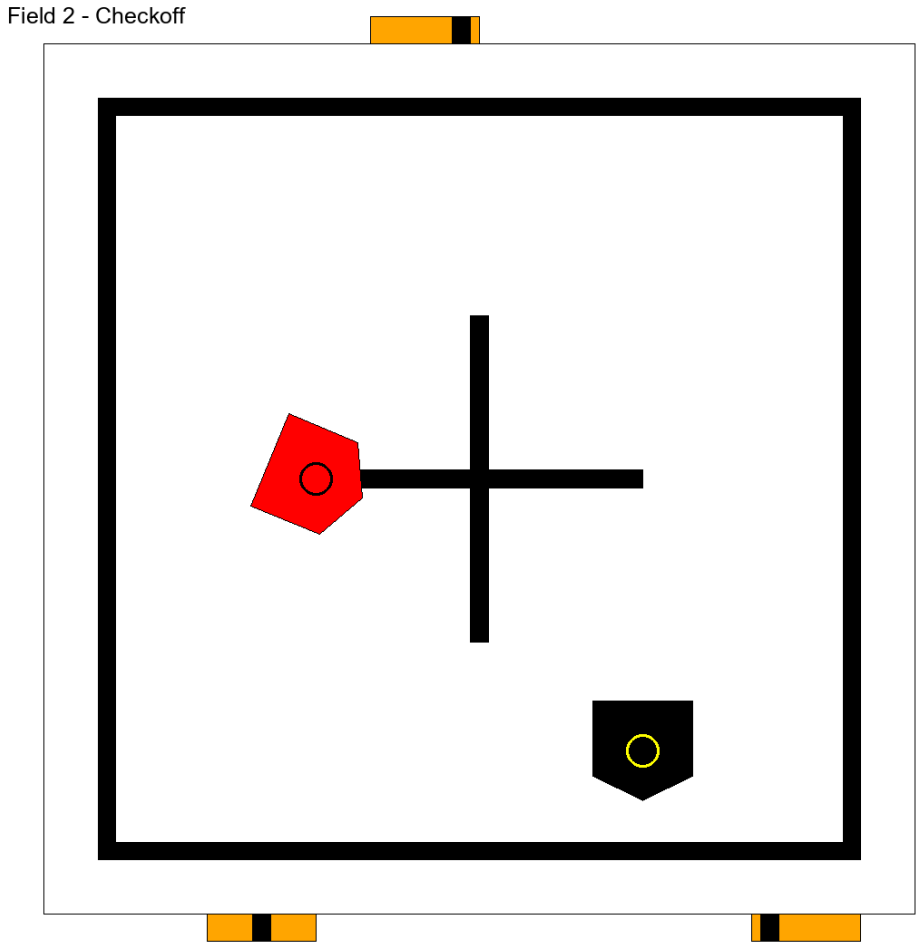


Figure 3: Example minimum-specification checkoff configuration. The red robot starts near the center in a random orientation; the stationary opponent (dead robot) is shown in black. Three slime tower targets (orange) sit outside the field, with black stripes indicating the marked hole positions. The robot must score two direct hits on the opponent and put at least one ball through the marked hole in each of at least two different towers.

At the start of every round, your robot must contain all ping-pong balls that it will use; no reloading is allowed. Both robots' 2 KHz IR beacons and all the slime tower targets' track wires remain active throughout the round. The correct tower hole is marked by a vertical banner.

[§]Yes, we have had teams go well into winter break before they finished. No one passes the class without passing min-spec.
[¶]Alternately, you could do some work on your other classes and not actually fail them.

Min-spec requires the following within a single 2-minute round:

1. Hit the opponent (dead) robot body twice with ping-pong balls. Hits must be direct, rather than ricochets off the field or other objects.
2. Put at least one ping-pong ball through the hole marked by the black vertical stripe in at least two of the slime tower targets.

Both requirements must be met; neither task substitutes for the other. A ball that hits the tower without passing through the designated hole, or passes through a different hole, does not satisfy the tower requirement. The two tasks may be completed in any order.[‖]

Each round allows up to 2 minutes to complete the challenge. Should it become apparent that a robot cannot finish, for example if it fails to break contact after a collision within 5 seconds or has insufficient ammunition remaining to complete the tasks, the robot will be disqualified and the round will end.

Tournament

In the tournament, you play against another team. Each robot starts near the center of its field in a random orientation and must locate its opponent and the slime tower targets. The objectives are the same as for min-spec: hit the opponent robot body twice and put at least one ball through the target hole marked by the vertical banner on two different towers.

Public tournament winners are determined by points. The winning robot advances in the tournament. Should a tie occur, we will attempt a rematch.[**]

Points are awarded as follows. Tower scores require a ping-pong ball to pass through the hole marked by the black vertical stripe. Note that the first bot to complete the full task wins regardless of points, but points will be used to determine the winner in the event of a tie.

Points	**Scoring event**
20	First direct hit to the opponent robot body (once per round).
40	Second direct hit to the opponent robot body (once per round).
10	Each subsequent direct hit to the opponent robot body.
50	First ball through the marked hole in a slime tower target (once per round).
100	Second ball through the marked hole in a different slime tower target (once per round).
150	Third ball through the marked hole in the remaining slime tower target.
10	Each subsequent ball through a marked hole in any slime tower target, awarded only after at least two different towers have been scored.

Tournament points do not replace the min-spec requirements for grading: completing the challenge requires both two opponent hits and at least one ball through the designated tower hole for at least two towers in the same round.[††]

Robots will be disqualified for going out of bounds, defined as more than half the robot over the black tape boundary, or for failing to break contact after a collision within 5 seconds.

We may, and almost certainly will, update these rules and/or points should flaws become apparent.

[‖] That is, we don't care if you go for a slime tower, then the dead bot, then the other slime tower, or any combination thereof. We just want to see two hits on the dead bot, and two balls through the right hole on two different towers.

[**] With 23 teams and two minute rounds, we are looking at over an hour and a half with no setup time in between matches.

[††] However, if you manage to complete min-spec during the tournament competition, we will certainly honor it.

The Droid

The droid must be a stand-alone entity that fits in an 11 in. x 11 in. x 11 in. cube at the beginning of the round. Your machine must contain all ping-pong ball ammo at the beginning of the round; no reloading is allowed. It should be capable of meeting all specifications while drawing power only from batteries. It must be able to detect bumps at a height of 3.0 inches above the ground. The droid must be able to detect and resolve collisions with an obstacle by breaking contact within 5 seconds. Droids should be able to keep themselves on the field, with at least half of the robot within the black 2 in. tape boundaries.

Droids will be programmed in C, using the standard MPLAB X IDE. Your droid behavior will be constructed using the ES_Framework from Lab 0. However, you may not use the Roach projects you wrote.[‡‡] You may reprogram your droid between rounds if you desire, but you may not alter it once the field configuration is established.

Each droid will be equipped with a remote power switch using the remote switch header on the Uno32 stack. You will not pass min-spec if you do not have a working remote power switch. At the beginning of the round, you will switch on your droid, and may not interfere with it until the round ends.

Note that all ping-pong balls must be launched from the droid itself; The balls may not be attached to the droid in any way. The balls must pass through the holes in the slime tower targets to count for scoring or min-spec.

Materials

Each team will be provided with one Uno Stack, one H-Bridge, one Stepper Board if needed, one DS3658 board, one battery, and one ULN2003 (if needed). BELS will provide two gearhead DC motors, rollerblade wheels, RC servos, wood, gluesticks, etc. There will also be wire, regulators, and solder freely available in the ECE-118 labs.

Each team should not exceed a budget of $150 total for other parts on the robot, and must maintain an up-to-date bill of materials (BOM). If we spot a nice $5K gyro on your bot, we will hand you $150 and take it. We do not want the project to be an arms race over who can purchase better stuff. We will have wood and foamcore available for purchase at cost. We will maintain a list of Amazon suggestions for motors, perf board, and other components within the first week of the project. BELS, Ace Hardware, Fastenal, and Home Depot are all decent local sources. HSC and Tap Plastics are excellent resources in the Bay Area. McMaster-Carr will deliver nearly any piece of hardware within a couple days but tends to be expensive. Amazon Prime is free to students for a trial period, and can get things to you quickly.

Available Tools

It should go unsaid that all work needs to be done by the team and not contracted out. You will have the resources in BE111, BE115, and BE150 as well as the drill press, tool chest, and laser cutter in BE138. Your circuits must be soldered on perfboards; no breadboards. Those of you thinking about PCB houses will not get turnaround in time without blowing your $150 budget. Off-the-shelf sensor boards, such as those sold by SparkFun or Adafruit, are fine, but understand that software integration for these sensors can take much longer than you anticipate. Manage your time carefully.

Field Specifications

We will have an OnShape model of the field, slime tower targets, and robot beacon mounts available after the midterm. The field layout in Fig. 1 will be available on the website in higher resolution and is drawn to scale. The target drawings will specify hole dimensions, the black vertical stripe, track wire routing, and

[‡‡] If we see a Roach_HSM file we will delete all of your code. Seriously, write it from scratch.

target placement; the field layout alone does not define these details. Modifications, if any, to the field will be noted on Ed Discussions.

Further questions or clarifications about the field specifications should be posted to the Ed Discussions forum.

Safety

The machines should be safe to the user, the lab, and the spectators. For this project, excessively high velocity ball delivery will be discouraged, so go ahead and forget about that CO_2 PVC pipe launcher you were thinking about. Voltages are limited to the rechargeable batteries in the lab. You may purchase your own if you would like, but consider 10 V an upper limit. Intentional jamming or blocking of the opposing robot, or masking of any beacon or track wire, is considered foul play and is not allowed. Robots deemed unsafe will be disqualified.

Young children line the competition field; take this into consideration when designing your launch mechanisms. Each team will be required to take three ping-pong ball shots from their own robot on bare flesh at a distance of 3 ft. from the barrel of their robot. All members of the team must do this.[§§]

Prior to competition your robot should not transcend space or time in any way, nor should your robot alter gravity within our Solar System. However, during competition, gravity and space-time may be altered at will.

Evaluation

Performance testing procedures: At least one team member will set up and start each machine; the robot must then complete the round autonomously. There will be one round for grading purposes done in the lab to evaluate robot performance on both tasks: two direct hits to the opponent robot body and at least one ball through the designated slime tower hole. The public tournament is purely for entertainment purposes, though if you have not yet checked off, successful completion of the min-spec tasks during the public demo counts as a valid late checkoff.

Grading evaluation: Each machine will be graded based on its performance in the testing before the class competition at the end of the quarter. Each machine will have up to 2 minutes to solve the challenge. Grading is not based on point values, but on how robustly your robot performs.

Grading Criteria

- **Concept (20%):** This will be based on the technical merit of the design and coding for the machine. Included in this grade will be evaluation of the appropriateness of the solution, as well as innovative hardware and software and use of physical principles in the solution.
- **Implementation (20%):** This will be based on the prototype displayed at the evaluation session. Included in this grade will be an evaluation of the physical appearance of the prototype and the quality of construction. We will not presume to judge true aesthetics, though we might comment on it, but will concentrate on craftsmanship and finished appearance.
- **Report (10%):** This will be based on an evaluation of the written report. It will be judged on clarity of explanations, completeness, and appropriateness of the documentation.
- **Performance (20%):** Based on the results of the performance during the evaluation session.
- **Design Evaluations (30%):** Based on check-off completion.

[§§] Any welts will be photographed and displayed prominently on the slideshow during the Public Demo

Project Milestones

Each week, your team will need to achieve a list of check-offs to stay on schedule and each partner will need to work as part of the team. If you do not stay on schedule with the check-offs, you will not finish in time and will be forced to stay through finals week until your robot is complete: stay on schedule.

Your weeks will essentially break into the following:

- **Week 1:** Design, schedule, group order, and design downselect. Preliminary Design Review.
- **Week 2:** Electronics and mechanical prototyping.
- **Week 3:** Working prototype for moving robot and ball launcher; state machine. Mid-Project Review.
- **Week 4:** Finalizing robot and getting everything to work together.
- **Week 5:** Competition and Final Check-Off.

There will be weekly checkoffs, three design reviews throughout the project, one lab report, and one and only one competition.

Half of this project is communicating well and documenting progress to stay on schedule. With that in mind, we expect each team to maintain and update a lab notebook with everything you are doing and copious notes. We very strongly recommend that your lab notebook stays with the robot in lab. We will use this to verify your check-offs for each week. We recommend sharing some form of file drive, team drive, or Git repository to help you keep yourselves on task, but do not require it. That said, each team will need to submit their lab notebook and schedules for the Preliminary Design Review. If you want to use Git for your project storage, you will be able to create a repo on the UCSC GitLab server.

Each student will also submit a short confidential team status report each week. This is not intended to create drama or assign blame; it is an early-warning system. If one person is disappearing, two people are doing all the work, nobody knows what anyone else is doing, or the team is politely driving off a cliff together, we want to know while there is still time to help.

A report describing the technical details of the machine will be required. The report should be of sufficient detail that any skilled ECE-118 alum could understand, reproduce, and modify the design.[¶¶]

Design Review 1: Preliminary Design Review

Date: Thursday, 05-Nov-2026.

Team Concepts: present your best design to the class for three minutes.

Come up with 3 team concepts for your design from your individual ideas and a bit of brainstorming. You will present your best 2. Mix and match between the best of your designs. How are you and your team going to accomplish your project goals? Schedule out your time as well as your team's.

You will need to upload your designs to Canvas, and present your lab notebook to the tutors/TAs for checkoff. Submit your best team design and backup design before class on Thursday. You will have 3 minutes to present your design and get some feedback on it. Have a primary and a backup in case it is too similar to someone else's.

Design Review 2: Mid-Project Review

Date: Wednesday, 25-Nov-2026.

Full Prototype: presented to the staff for 15 minutes.

[¶¶]This report will be one of your portfolio pieces and is very useful in either graduate school or industry applications.

Present your currently working parts and your full design to the instructors for review and insight into potential roadblocks. Every system, both mechanical and electrical, should be prototyped at this point.

Mechatronics Beer Challenge

Each team gets exactly three consecutive tries on the field to successfully complete the final check-off. If you can complete the task 3 out of 3 attempts, and your robot still functions by meeting min spec in the public demo, you get beer. In the history of mechatronics only a few teams have succeeded, and it was easier then. Note that during the beer challenge, the field is not random, but possibly set in a way to be difficult for your particular robot. *We reserve the right to be evil here*.

Design Review 3: Final Check-Off

Date: Wednesday, 02-Dec-2026.

Present your final check-off robot to the staff. You get three tries to succeed on the field in each session.

Deliverables are:

- Robot that starts near the center in a random orientation and autonomously completes both tasks in a single round: two direct opponent body hits on the dead robot and at least one ball through the designated slime tower hole on two different towers.

Competition: Friday, 04-Dec-2026. The public demo of your finished, operational machines is a fun performance and will likely have a large and enthusiastic audience. It will be held in a room to be announced, starting at 6:00 PM. You will be expected to arrive early for setup.

There will be a post-tournament decompression at one of the Santa Cruz watering holes.[|||] We will post plans on Ed Discussion.

Clean-up and Class Review: Monday after the Public Demo.

Lab Report

Due: Thursday, 10-Dec-2026.

Electronic copy of your lab report, turned in as a group assignment in Canvas.

Create a section for each design and write an evaluation of each aspect of your design: what went well and what did not. Make sure to include pictures and links to video as necessary. Also include your final BOM.

Check-Off Schedule

Check-offs are used to ensure that you are on track and keeping up. They allow the teaching staff to allocate their resources to help teams out in an appropriate way. Treat them seriously and your project will be completed well and on time.

For each check-off after teams are assigned, each student will submit the Self- and Team Reflection form. This should take two minutes, not twenty. Tell us what you personally contributed since the last checkoff, what you are responsible for next, whether the team is meeting and communicating effectively, and whether there is anything the staff should know before it becomes a train wreck. A completely normal report saying "things are fine" is perfectly acceptable. A report saying "things are not fine" is much more useful before the team is screaming at each other.[***]

[|||] Typically Woodstock Pizza, as they have an outdoor area and counter service. Friends and Family are more than welcome to join.

[***] You are going to be under time pressure, and working with people who you do not necessarily know that well (though you will by the end of the project). There are going to be personality conflicts, scheduling conflicts, and style conflicts. A small intervention early is much easier than a big intervention late—let us know things are going wrong sooner rather than later.

Check-off 1: Team Designs

Date: Wednesday, 04-Nov-2026.

Using the three concepts that each of you created for the midterm, now that you are assigned to teams, come up with 3 team concepts for your design. Mix and match between the best of your individual designs. How are you and your team going to accomplish your project goals? Get some details written down in your engineering notebook.

Deliverables are:

- 3 detailed team concepts for solving the project.
- Make copies of them for your teammates and yourself.

Check-off 2: Design Downselect and Team Health

Date: Sunday, 08-Nov-2026.

Before the project disappears into the long gap between the first team designs and the full team assignment checkoff, each team must show that they have actually started working as a team. This is the point where you should stop admiring all possible robots and pick the one you are really going to build.

Deliverables are:

- Primary design selected, with a short explanation of why this design won.
- Backup design or fallback path if the primary launcher, drive, or sensing approach fails.
- Initial list of parts that must be ordered, borrowed, fabricated, or tested immediately.
- At least one concrete project task completed by each team member since Check-off 1.
- Individual confidential team status reports submitted by all three team members.

Check-off 3: Team Assignments, Tasks, Working Beacon and Track Wire

Date: Thursday, 12-Nov-2026.

Basic project management and system component design. This is where you define who is primary and secondary on which tasks, how you will coordinate time and schedules, and how the major systems fit together.

Deliverables are:

- Time schedules.
- Personnel assignments.
- Primary and secondary owners for each major subsystem.
- System block diagram.
- Mechanical design sketches.
- Working beacon detectors on perf-board, at least one, with LEDs and accompanying schematics.[†††]
- Working track wire detector for identifying the slime tower targets, with an indicator and schematic.

Check-off 4: Final CAD and State Machine Drawings

Date: Monday, 16-Nov-2026.

Mechanical and software designs. You should at this point have your final robot design completed in appropriate CAD software (OnShape), and your state machines should be entirely drawn out neatly. Your mechanical design should easily fit within an 11 in. cube. While both of these may need to be updated as you progress, they should be close to final form.

Deliverables are:

[†††]This should be the best one from Lab 2 among your team. If you really, truly, have to build a new one, make sure you use the best of the designs from the lab. At this point, you do not want to redo lab 1 and lab 2, just replicate one if your working designs.

- State machine or state machines.
- Final mechanical design in OnShape.

Check-off 5: Prototyped Sensors and Actuators

Date: Friday, 20-Nov-2026.

Sensors and actuators. Your full sensor suite should be functional and documented at this point, including opponent beacon detection, target track wire detection, black vertical stripe detection, boundary sensing, and collision sensing. All ball launch mechanisms should be prototyped and tested for range and accuracy against both the opponent robot body and the slime target tower holes. This gives time to redesign if necessary.

Deliverables are:

- Working sensors, breadboard is okay, and schematics.
- Actuators, breadboard is okay, and schematics.

Check-off 6: Final Sensors and Actuators

Date: Wednesday, 25-Nov-2026.[‡‡‡]

Final sensor and actuator designs. Final working perfboard prototypes of all sensors and actuators that your robot will use. Fully tested, fully documented.

Deliverables are:

- Final sensors and final schematics.
- Final actuators and final schematics.

Check-off 7: Basic Navigation

Date: Monday, 30-Nov-2026.

Mobile platform with basic reactive navigation. Starting near the center in a random orientation, your robot should be able to find its bearings, keep itself on the field with tape sensing, and resolve collisions with bump sensors.

Deliverables are:

- Autonomous platform that can move and sense.
- Reverse off of a collision sensor.
- Keep itself on the field.

Check-off 8: Full Navigation

Date: Wednesday, 02-Dec-2026.

Robot that can start near the center in a random orientation, locate the opponent using its beacon, locate the slime tower target using track wire, and identify the correct hole using the black vertical stripe. Integrate navigation, aiming, and ball launching for both tasks.

Deliverables are:

- Robot that autonomously navigates and scores two direct hits to the opponent robot body.
- Robot that identifies the designated target and marked hole, then puts at least one ball through that hole in the same round.

[‡‡‡]Note that this is the last class day before Thanksgiving holiday. Give a serious evaluation with your teammates and decide if you need the extra days to get your robot working, or if you can take it easy and enjoy the Turkey. Have the optional extra time is an advantage to taking ECE118 in the Fall quarter.

- Robot fits entirely within the Cube of Compliance at the start of the round.

Specifications Check-Off: Min Spec

Date: Friday, 04-Dec-2026.

Your robot should be able to meet the minimum specifications. After that you may go home and celebrate, or sleep.

Deliverables are:

- Robot that autonomously completes both tasks within one 2-minute round from a random orientation near the field center: two direct hits to the opponent robot body and at least one ball through the black-stripe-marked hole in the trackwire-designated slime tower target.

Notes on Successful Project Management

There are a few rules of thumb to follow that will make your project much more successful, and keep you working well as a team.

The first rule is a bit paradoxical, but nonetheless important: *do what you are bad at*. That is, if you are good at software but bad at mechanics, then you take the lead on mechanical stuff and take a secondary role in software design.

The second rule: double-team every single task you need done. That means one person is primary or lead, and the other is secondary. Note that if you follow the first tip, then likely the secondary is better at the task than the primary. Do not attempt to split tasks up so that each one of you goes off and does it and then comes back. This never works and is always slower in the long run.

When crunch time comes, you can run a rotation with your three team members such that one sleeps and two work. The just-woken-up one works under the one who has been up. Then the lead goes to sleep, the secondary goes into lead position on another task, and the sleeping one gets woken up to be secondary. While this is not sustainable beyond a couple of weeks, you can get an enormous amount done this way.[§§§]

Be careful about sleeplessness and cars, bikes, and so on. There are plenty of couches around to crash on, and a number of students live in GSH, 200 ft. from the lab. Do not think you can keep yourself awake long enough to drive or bike home. Be smart about this. We really do not want to see anyone get hurt through a senseless crash.[17]

With this many people in the lab, it is going to be very important that you keep the lab clean and not leave your things lying around. We will be assigning I/O boards and batteries to each team, and they will be yours until the project is over. Don't snag other groups' batteries or equipment. If you need to borrow something, ask the other team first. If you need something fixed or replaced, let the staff know.

People occasionally donate random parts, and if you happen to find surplus printers or other random electronics that people no longer want, feel free to dismantle and salvage what you want. However, please discard all parts that are not salvageable in an appropriate e-waste container so as to reduce clutter in the lab.

Drive motors have, in general, been a make-or-break part of the project. BELS will hand out some decent ones, but if you need something different, I would strongly suggest you consider purchasing some gearhead motors from Jameco, MPJA.com, or Amazon.com. Ordering them early, meaning now, would ensure that you have a set that will work by the time you need them. I will post on Ed Discussion what I think are decent motors. If you have Prime, they will get here quickly.

[§§§] I personally have kept this up for approximately 21 days—not recommended but boy did we get a lot done (DoD deadline)

[17] If you need a ride home and are too sleepy, call one of the staff—we will come get you or find a couch for you.

Mechatronics Beer Challenge Reminder

Any team that is able to complete the beer challenge spec with a fully functioning and finished droid a full week ahead of the deadline gets a case of beer or other equivalent adult beverage, within reason, supplied by The Prof. Only a few teams have ever collected this. See rules above.